

AMS 580 Quiz 3

Kachun Huang

2025-03-10

AMS 580 Quiz 3

Import Library

```
library(neuralnet)
```

```
## Warning: package 'neuralnet' was built under R version 4.3.3
```

```
library(MASS)
library(tidyverse)
```

```
## Warning: package 'purrr' was built under R version 4.3.3
```

```
## Warning: package 'lubridate' was built under R version 4.3.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::compute() masks neuralnet::compute()
```

```
## x dplyr::filter()  masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## x dplyr::select()  masks MASS::select()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
## Warning: package 'lattice' was built under R version 4.3.3
```

```
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
## lift
```

```
library(dplyr)
```

Task 1

Data Import

```
data("airquality")
data <- airquality
head(data)
```

```
##   Ozone Solar.R Wind Temp Month Day
## 1    41     190  7.4   67     5   1
## 2    36     118  8.0   72     5   2
## 3    12     149 12.6   74     5   3
## 4    18     313 11.5   62     5   4
## 5     NA      NA 14.3   56     5   5
## 6    28      NA 14.9   66     5   6
```

```
str(data)
```

```
## 'data.frame':   153 obs. of  6 variables:
## $ Ozone   : int  41 36 12 18 NA 28 23 19 8 NA ...
## $ Solar.R: int  190 118 149 313 NA NA 299 99 19 194 ...
## $ Wind    : num  7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
## $ Temp    : int  67 72 74 62 56 66 65 59 61 69 ...
## $ Month   : int  5 5 5 5 5 5 5 5 5 5 ...
## $ Day     : int  1 2 3 4 5 6 7 8 9 10 ...
```

Data Pre-processing

remove missing values

```
data <- na.omit(data)
head(data)
```

```
##   Ozone Solar.R Wind Temp Month Day
## 1    41     190  7.4   67     5   1
## 2    36     118  8.0   72     5   2
## 3    12     149 12.6   74     5   3
## 4    18     313 11.5   62     5   4
## 7    23     299  8.6   65     5   7
## 8    19      99 13.8   59     5   8
```

```
str(data)
```

```
## 'data.frame': 111 obs. of 6 variables:
## $ Ozone : int 41 36 12 18 23 19 8 16 11 14 ...
## $ Solar.R: int 190 118 149 313 299 99 19 256 290 274 ...
## $ Wind : num 7.4 8 12.6 11.5 8.6 13.8 20.1 9.7 9.2 10.9 ...
## $ Temp : int 67 72 74 62 65 59 61 69 66 68 ...
## $ Month : int 5 5 5 5 5 5 5 5 5 5 ...
## $ Day : int 1 2 3 4 7 8 9 12 13 14 ...
## - attr(*, "na.action")= 'omit' Named int [1:42] 5 6 10 11 25 26 27 32 33 34 ...
## ..- attr(*, "names")= chr [1:42] "5" "6" "10" "11" ...
```

After removing missing values, there are 111 observations left.

Compute mean and std

```
mean <- mean(data$Ozone)
print(mean)
```

```
## [1] 42.0991
```

```
sd <- sd(data$Ozone)
print(sd)
```

```
## [1] 33.27597
```

Scale the data

```
data <- data.frame(scale(data))
head(data)
```

```
##      Ozone      Solar.R      Wind      Temp      Month      Day
## 1 -0.03302982  0.05702761 -0.7138405 -1.1325108 -1.504117 -1.7165054
## 2 -0.18328840 -0.73285918 -0.5451928 -0.6078501 -1.504117 -1.6016578
## 3 -0.90452961 -0.39276904  0.7477726 -0.3979858 -1.504117 -1.4868103
## 4 -0.72421931  1.40641756  0.4385852 -1.6571715 -1.504117 -1.3719627
## 7 -0.57396073  1.25282846 -0.3765451 -1.3423751 -1.504117 -1.0274200
## 8 -0.69416759 -0.94130153  1.0850679 -1.9719679 -1.504117 -0.9125725
```

```
str(data)
```

```
## 'data.frame': 111 obs. of 6 variables:
## $ Ozone : num -0.033 -0.183 -0.905 -0.724 -0.574 ...
## $ Solar.R: num 0.057 -0.733 -0.393 1.406 1.253 ...
## $ Wind : num -0.714 -0.545 0.748 0.439 -0.377 ...
## $ Temp : num -1.133 -0.608 -0.398 -1.657 -1.342 ...
## $ Month : num -1.5 -1.5 -1.5 -1.5 -1.5 ...
## $ Day : num -1.72 -1.6 -1.49 -1.37 -1.03 ...
```

Data Splitting

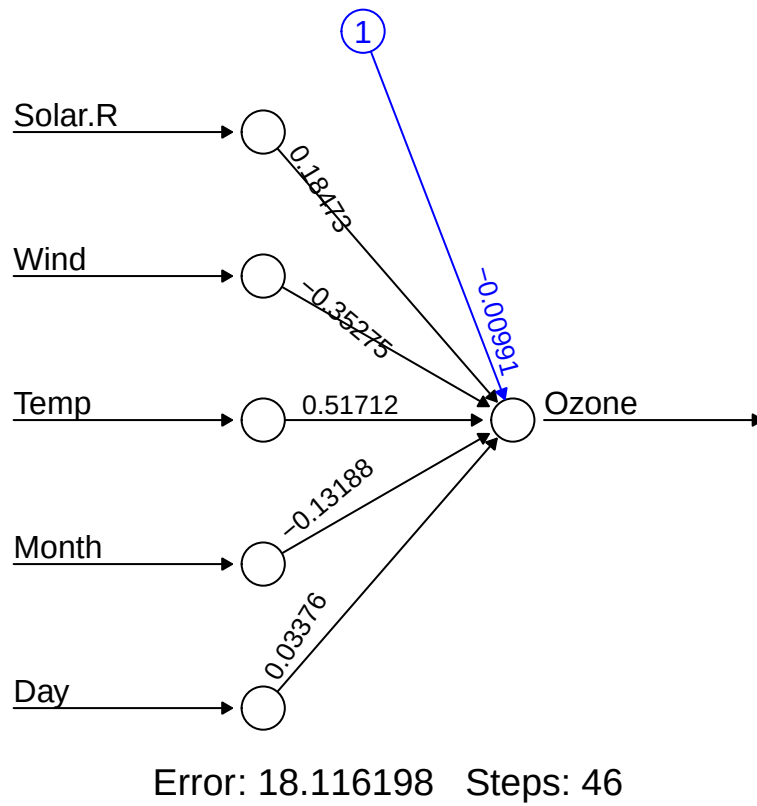
```
set.seed(123)
training.sample <- data$Ozone %>% createDataPartition(p=0.8, list=FALSE)
train.data <- data[training.sample,]
test.data <- data[-training.sample,]
```

NN with no hidden layer

```
nn_no_model <- neuralnet(
  Ozone ~ .,
  train.data,
  hidden = 0,
  err.fct = "sse",
  linear.output = TRUE
)
```

plot

```
plot(nn_no_model, rep="best")
```



Predict on Test Data

```
pred_nn_no <- predict(nn_no_model, test.data)
```

Descale

```
# compute y(unscaled)
y <- sd * test.data$Ozone + mean
#compute y^hat (unscaled)
yhat <- pred_nn_no*sd+mean
```

Compute Test MSE

```
sum((y-yhat)^2)/length(y)
```

```
## [1] 286.2053
```

or

```
(MSE.nn.0 = RMSE(y,yhat)^2)
```

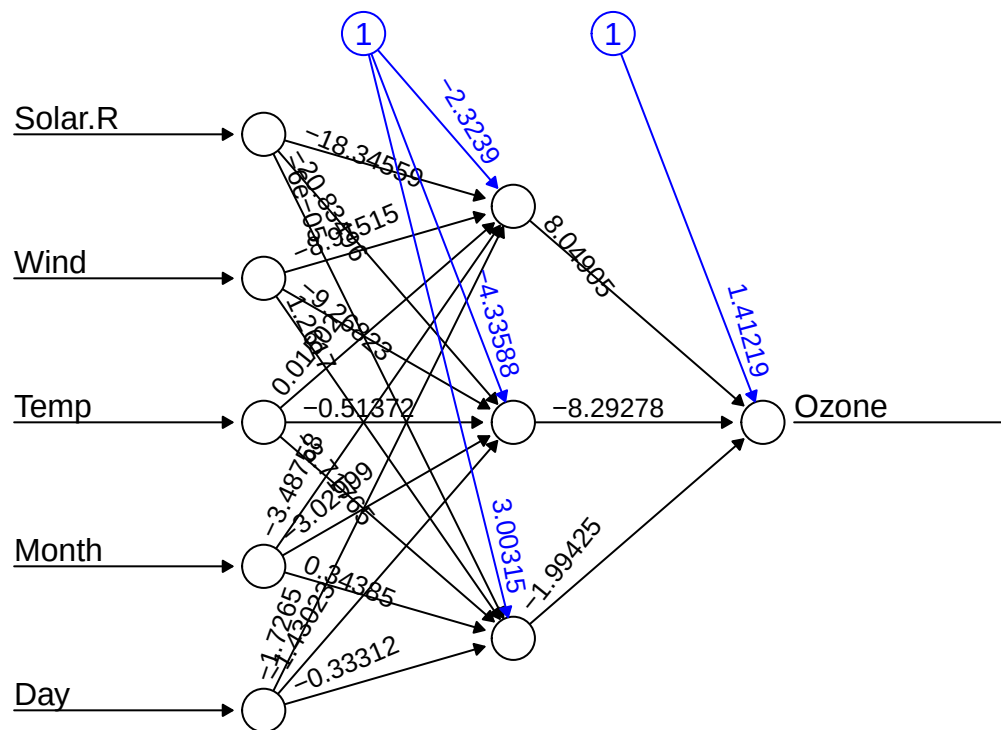
```
## [1] 286.2053
```

NN with one hidden layer

```
nn_one_model <- neuralnet(
  Ozone ~.,
  train.data,
  hidden = 3,
  err.fct = "sse",
  linear.output = TRUE
)
```

plot

```
plot(nn_one_model, rep="best")
```



Error: 4.7173 Steps: 7020

Predict on Test Data

```
pred_nn_one <- predict(nn_one_model, test.data)
```

Descale

```
# compute y(unscaled)
y <- sd * test.data$Ozone + mean

#compute y^hat (unscaled)
yhat <- pred_nn_one*sd+mean
```

Compute Test MSE

```
sum((y-yhat)^2)/length(y)
```

```
## [1] 868.5274
```

or

```
(MSE.nn.1 = RMSE(y,yhat)^2)
```

```
## [1] 868.5274
```

Result

The model with no hidden layer is better than the model with one hidden layer by test MSE. The possible reason could be the relationship is closer to linear so adding extra layer which introduce nonlinear could either overfit or it doesn't fit well.

Task 2

Data Import

```
titantic <- read.table("/Users/thomashuang/Downloads/Titanic2.csv", header=TRUE, sep=",")
head(titantic)
```

```
## PassengerId Survived Pclass
## 1          1         0       3
## 2          2         1       1
## 3          3         1       3
## 4          4         1       1
## 5          5         0       3
## 6          6         0       3
##
##              Name      Sex Age SibSp Parch
## 1              Braund, Mr. Owen Harris    male  22      1      0
## 2 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female  38      1      0
## 3              Heikkinen, Miss. Laina female  26      0      0
## 4 Futrelle, Mrs. Jacques Heath (Lily May Peel) female  35      1      0
## 5              Allen, Mr. William Henry    male  35      0      0
## 6              Moran, Mr. James      male  NA      0      0
##
##      Ticket      Fare Cabin Embarked
## 1      A/5 21171  7.2500      S
## 2      PC 17599 71.2833    C85      C
## 3 STON/O2. 3101282  7.9250      S
## 4      113803 53.1000   C123      S
## 5      373450  8.0500      S
## 6      330877  8.4583      Q
```

```
str(titantic)
```

```
## 'data.frame':    891 obs. of  12 variables:
## $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
## $ Survived   : int  0 1 1 1 0 0 0 0 1 1 ...
## $ Pclass     : int  3 1 3 1 3 3 1 3 3 2 ...
## $ Name       : chr  "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
## $ Sex        : chr  "male" "female" "female" "female" ...
## $ Age        : num  22 38 26 35 35 NA 54 2 27 14 ...
```

```
## $ SibSp      : int  1 1 0 1 0 0 0 3 0 1 ...
## $ Parch      : int  0 0 0 0 0 0 0 1 2 0 ...
## $ Ticket     : chr   "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
## $ Fare       : num   7.25 71.28 7.92 53.1 8.05 ...
## $ Cabin      : chr    "" "C85" "" "C123" ...
## $ Embarked   : chr    "S" "C" "S" "S" ...
```

Data Pre-processing

Subset

```
titantic <- subset(titantic, select = c(Pclass, Sex, Age, Survived))
head(titantic)
```

```
##   Pclass   Sex Age Survived
## 1      3  male  22         0
## 2      1 female  38         1
## 3      3 female  26         1
## 4      1 female  35         1
## 5      3  male  35         0
## 6      3  male  NA         0
```

```
str(titantic)
```

```
## 'data.frame':   891 obs. of  4 variables:
## $ Pclass   : int   3 1 3 1 3 3 1 3 3 2 ...
## $ Sex      : chr    "male" "female" "female" "female" ...
## $ Age      : num   22 38 26 35 35 NA 54 2 27 14 ...
## $ Survived: int    0 1 1 1 0 0 0 0 1 1 ...
```

remove missing values

```
titantic <- na.omit(titantic)
head(titantic)
```

```
##   Pclass   Sex Age Survived
## 1      3  male  22         0
## 2      1 female  38         1
## 3      3 female  26         1
## 4      1 female  35         1
## 5      3  male  35         0
## 7      1  male  54         0
```

```
str(titantic)
```

```
## 'data.frame':   714 obs. of  4 variables:
## $ Pclass   : int   3 1 3 1 3 1 3 3 2 3 ...
## $ Sex      : chr    "male" "female" "female" "female" ...
```

```
## $ Age      : num  22 38 26 35 35 54 2 27 14 4 ...
## $ Survived: int   0 1 1 1 0 0 0 1 1 1 ...
## - attr(*, "na.action")= 'omit' Named int [1:177] 6 18 20 27 29 30 32 33 37 43 ...
## ..- attr(*, "names")= chr [1:177] "6" "18" "20" "27" ...
```

After remove the missing values, there are 714 observations left.

Dummy Variable

```
titantic$Sex <- ifelse(titantic$Sex == 'male', 1, 0)
head(titantic)
```

```
##   Pclass Sex Age Survived
## 1      3   1  22         0
## 2      1   0  38         1
## 3      3   0  26         1
## 4      1   0  35         1
## 5      3   1  35         0
## 7      1   1  54         0
```

Data Splitting

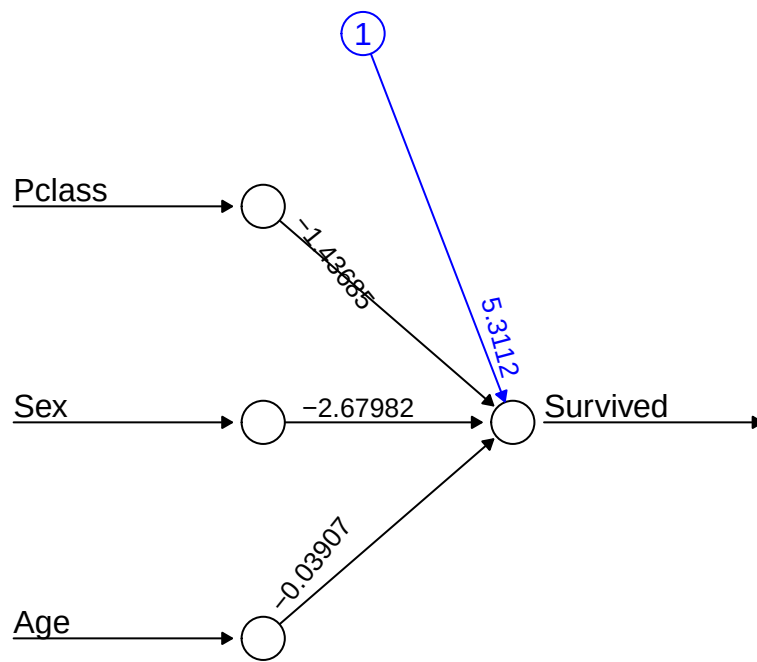
```
set.seed(123)
training.sample <- titantic$Survived %>% createDataPartition(p=0.8, list=FALSE)
train.data <- titantic[training.sample,]
test.data <- titantic[-training.sample,]
```

NN with no hidden layer

```
nn_no_model_c <- neuralnet(
  Survived ~.,
  train.data,
  hidden = 0,
  err.fct = "sse",
  linear.output = FALSE
)
```

plot

```
plot(nn_no_model_c, rep="best")
```



Error: 41.976268 Steps: 577

Predict on Test Data

```

pred_nn_no_c <- predict(nn_no_model_c, test.data)
pred_nn_no_c_class <- ifelse(pred_nn_no_c > 0.5,1,0)

```

Confusion Matrix

```

cm1 <- confusionMatrix(factor(pred_nn_no_c_class), factor(test.data$Survived), positive = "1")
cm1

```

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##           0 73 19
##           1  6 44
##
##           Accuracy : 0.8239
##           95% CI : (0.7512, 0.8827)
##           No Information Rate : 0.5563
##           P-Value [Acc > NIR] : 1.424e-11
##
##           Kappa : 0.6357
##
##           Mcnemar's Test P-Value : 0.0164
##

```

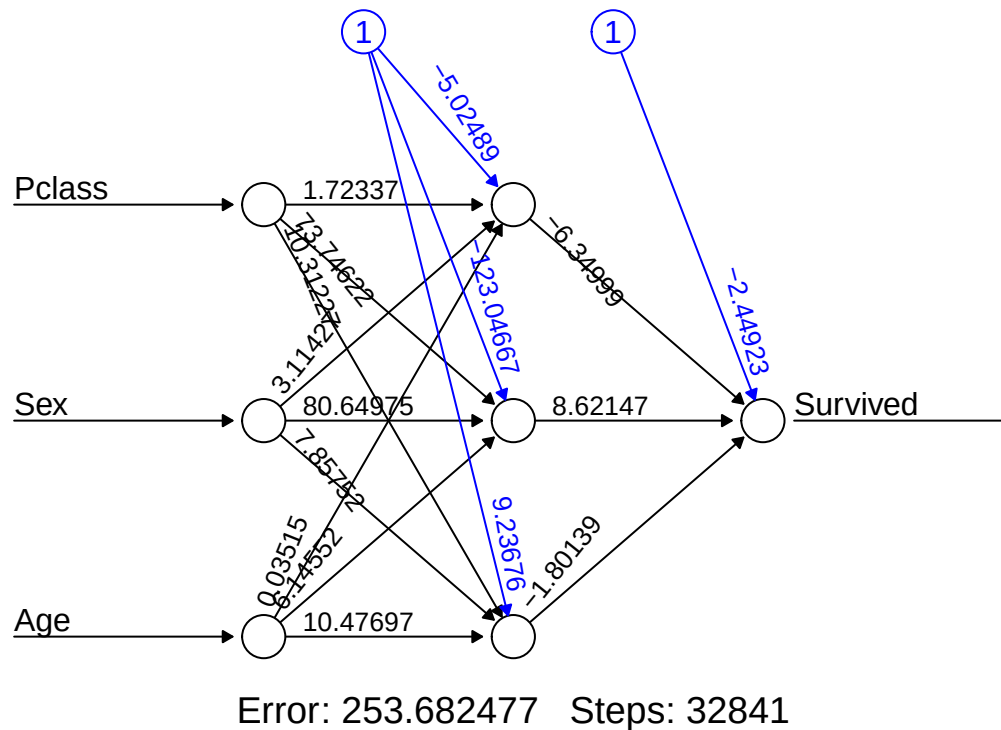
```
##          Sensitivity : 0.6984
##          Specificity : 0.9241
##          Pos Pred Value : 0.8800
##          Neg Pred Value : 0.7935
##          Prevalence : 0.4437
##          Detection Rate : 0.3099
##          Detection Prevalence : 0.3521
##          Balanced Accuracy : 0.8112
##
##          'Positive' Class : 1
##
```

NN with one hidden layer

```
nn_one_model_c <- neuralnet(
  Survived ~.,
  train.data,
  hidden = 3,
  err.fct = "ce",
  linear.output = FALSE
)
```

plot

```
plot(nn_one_model_c, rep="best")
```



Predict on Test Data

```
pred_nn_one_c <- predict(nn_one_model_c, test.data)
pred_nn_one_c_class <- ifelse(pred_nn_one_c > 0.5,1,0)
```

Confusion Matrix

```
cm2 <- confusionMatrix(factor(pred_nn_one_c_class), factor(test.data$Survived), positive = "1")
cm2
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##           0 75 22
##           1  4 41
##
##           Accuracy : 0.8169
##           95% CI : (0.7433, 0.8768)
##       No Information Rate : 0.5563
##       P-Value [Acc > NIR] : 5.201e-11
##
##           Kappa : 0.618
##
##  Mcnemar's Test P-Value : 0.0008561
##
##           Sensitivity : 0.6508
##           Specificity : 0.9494
##       Pos Pred Value : 0.9111
##       Neg Pred Value : 0.7732
##           Prevalence : 0.4437
##       Detection Rate : 0.2887
##   Detection Prevalence : 0.3169
##       Balanced Accuracy : 0.8001
##
##       'Positive' Class : 1
##
```

Result

- In terms of overall accuracy: model with no hidden layer is better.
- In terms of Sensitivity: model with no hidden layer is better.
- In terms of Specificity: model with one hidden layer is better.